

2025-10-30 - Valentini (9)

Previous lecture: [2025-10-24 - Servidio \(8\)](#)

Next lecture: [2025-10-31 - Valentini \(10\)](#)

Recap

$$D(k, \omega) = 1 - \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \int_L \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega/k} du = 0$$

$$F_{\alpha 0}(u) = \int_{-\infty}^{\infty} f_{\alpha 0}(u, w_1, w_2) dw_1 dw_2$$

$$f_{\alpha 0}(u, w_1, w_2) = \frac{n_{\alpha 0} m_{\alpha}^{3/2}}{(2\pi k_B T_{\alpha 0})^{3/2}} \exp \left[-\frac{m_{\alpha}}{2k_B T_{\alpha 0}} (u^2 + w_1^2 + w_2^2) \right]$$

$$\Rightarrow F_{\alpha 0}(u) = \int f_{\alpha 0} dw_1 dw_2 = \frac{n_{\alpha 0} m_{\alpha}^{1/2}}{(2\pi k_B T_{\alpha 0})^{1/2}} \exp \left[-\frac{m_{\alpha}}{2k_B T_{\alpha 0}} u^2 \right]$$

$$\left| \frac{\omega_I}{\omega_R} \right| \ll 1 \Rightarrow \begin{cases} D_R(k, \omega_R) = 0 \Rightarrow \omega_R = \omega_R(k) \\ \omega_I = -\frac{D_I(k, \omega_R)}{\partial D_R / \partial \omega_R} \end{cases}$$

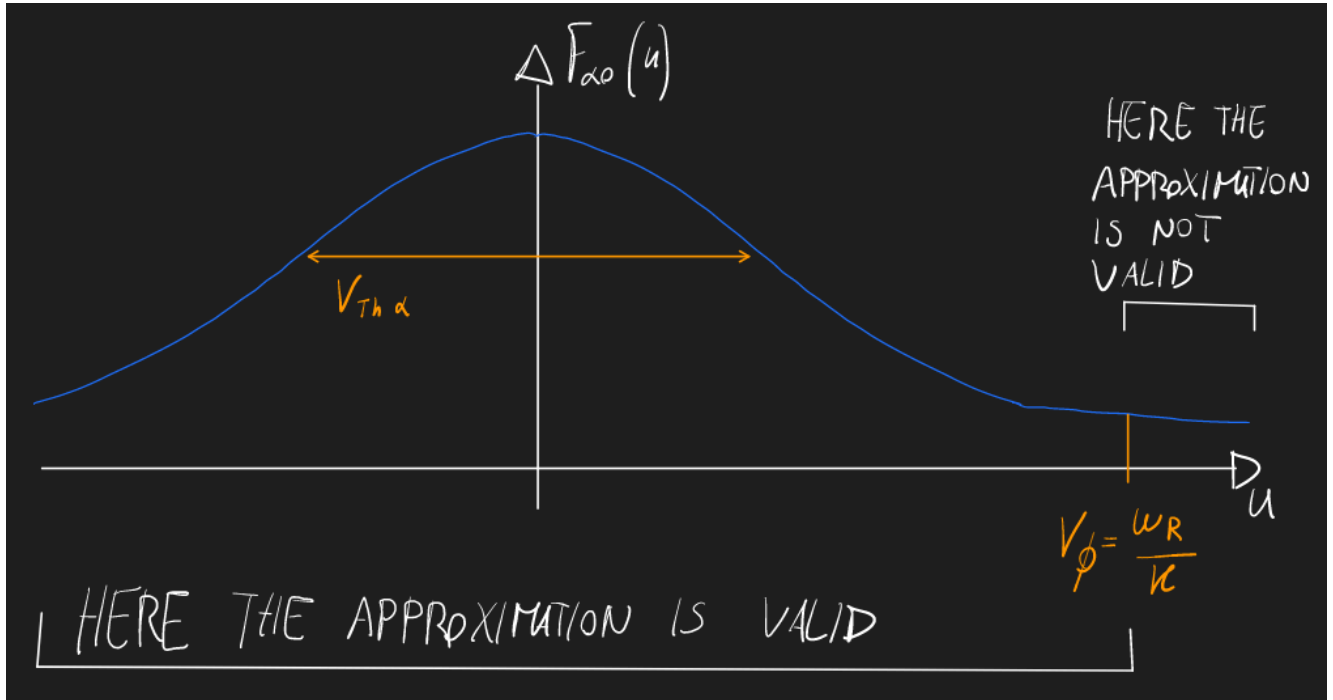
with

$$D_R = 1 - \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot P \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega_R/k} du$$

$$D_I = -\pi \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot \frac{\partial F_{\alpha 0}(u)}{\partial u} \Big|_{\omega_R/k}$$

High frequency regime

$$\frac{\omega_R}{k} \gg v_{th,\alpha}$$



The approximation is valid for the majority of the particles, except for a small fraction of particles in the tail of the distribution.

$$D_R = 1 - \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot P \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}(u)/\partial u}{u - \omega_R/k} du =$$

$$= 1 + \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot P \int_{-\infty}^{\infty} \frac{k}{\omega_R} \frac{\partial F_{\alpha 0}(u)/\partial u}{1 - \frac{uk}{\omega_R}} du$$

we expand the integral using Taylor series

$$P \int_{-\infty}^{\infty} \frac{k}{\omega_R} \frac{\partial F_{\alpha 0}(u)/\partial u}{1 - \frac{uk}{\omega_R}} du \approx \frac{k}{\omega_R} \int_{-\infty}^{\infty} \frac{\partial F_{\alpha 0}}{\partial u} \left[1 + \frac{uk}{\omega_R} + \frac{u^2 k^2}{\omega_R^2} + \frac{u^3 k^3}{\omega_R^3} + \dots \right] du = (*)$$

and we substitute $\partial F_{\alpha 0}/\partial u$ with the definition of $F_{\alpha 0}$ for a maxwellian distribution

$$(*) = \frac{k}{\omega_R} \int_{-\infty}^{\infty} \left[\frac{n_{\alpha 0} m_{\alpha}^{1/2}}{(2\pi k_B T_{\alpha 0})^{1/2}} \left(-\frac{m_{\alpha}}{k_B T_{\alpha 0}} u \right) \exp \left(-\frac{m_{\alpha}}{2k_B T_{\alpha 0}} u^2 \right) \right] \cdot$$

$$\cdot \left[1 + \frac{uk}{\omega_R} + \frac{u^2 k^2}{\omega_R^2} + \frac{u^3 k^3}{\omega_R^3} + \dots \right] du =$$

$$= -\frac{n_{\alpha 0} m_{\alpha}^{3/2}}{\sqrt{2\pi} (k_B T_{\alpha 0})^{3/2}} \int_{-\infty}^{\infty} \exp \left(-\frac{m_{\alpha}}{2k_B T_{\alpha 0}} u^2 \right) \cdot$$

$$\cdot \left[\cancel{\frac{uk}{\omega_R}} + \frac{u^2 k^2}{\omega_R^2} + \cancel{\frac{u^3 k^3}{\omega_R^3}} + \frac{u^4 k^4}{\omega_R^4} + \dots \right] du =$$

$$= -\frac{n_{\alpha 0} m_{\alpha}^{3/2}}{\sqrt{2\pi} (k_B T_{\alpha 0})^{3/2}} \left[\frac{k^2}{\omega_R^2} \int_{-\infty}^{\infty} u^2 \exp \left(-\frac{m_{\alpha}}{2k_B T_{\alpha 0}} u^2 \right) du + \right.$$

$$\left. + \frac{k^4}{\omega_R^4} \int_{-\infty}^{\infty} u^4 \exp \left(-\frac{m_{\alpha}}{2k_B T_{\alpha 0}} u^2 \right) du \right] = (*)$$

Where all the odd functions integrated in this (symmetric) integration interval gives zero as a result.

Now we change variable using

$$y^2 = u^2 \frac{m_\alpha}{2k_B T_{\alpha 0}}, \quad y = u \sqrt{\frac{m_\alpha}{2k_B T_{\alpha 0}}}$$

so we have

$$\begin{aligned} (*) &= -\frac{n_{\alpha 0} m_\alpha^{3/2}}{\sqrt{2\pi} (k_B T_{\alpha 0})^{3/2}} \left[\frac{k^2}{\omega_R^2} \int_{-\infty}^{\infty} y^2 \frac{2k_B T_{\alpha 0}}{m_\alpha} e^{-y^2} dy \sqrt{\frac{2k_B T_{\alpha 0}}{m_\alpha}} + \right. \\ &\quad \left. + \frac{k^4}{\omega_R^4} \int_{-\infty}^{\infty} y^4 \left(\frac{2k_B T_{\alpha 0}}{m_\alpha} \right)^2 e^{-y^2} dy \sqrt{\frac{2k_B T_{\alpha 0}}{m_\alpha}} \right] = \\ &= -\frac{n_{\alpha 0} \cancel{m_\alpha^{3/2}}}{\sqrt{2\pi} (\cancel{k_B T_{\alpha 0}})^{3/2}} \frac{k^2}{\omega_R^2} \left(\frac{2 \cancel{k_B T_{\alpha 0}}}{\cancel{m_\alpha}} \right)^{3/2} \left[\int_{-\infty}^{\infty} y^2 e^{-y^2} dy + \right. \\ &\quad \left. + \frac{k^2}{\omega_R^2} \frac{2k_B T_{\alpha 0}}{m_\alpha} \int_{-\infty}^{\infty} y^4 e^{-y^2} dy \right] = \\ &= -\frac{2n_{\alpha 0}}{\sqrt{\pi}} \frac{k^2}{\omega_R^2} \left[\frac{\sqrt{\pi}}{2} + \frac{k^2}{\omega_R^2} \frac{2k_B T_{\alpha 0}}{m_\alpha} \frac{3\sqrt{\pi}}{4} \right] = \\ &= -n_{\alpha 0} \frac{k^2}{\omega_R^2} \left[1 + 3 \frac{k^2}{\omega_R^2} \frac{k_B T_{\alpha 0}}{m_\alpha} \right] \end{aligned}$$

So we can substitute this result in

$$\begin{aligned} D_R &= 1 + \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot P \int_{-\infty}^{\infty} \frac{k}{\omega_R} \frac{\partial F_{\alpha 0}(u)/\partial u}{1 - \frac{uk}{\omega_R}} du = \\ &= 1 + \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{\cancel{k^2 n_{\alpha 0}}} \cdot \left\{ -\cancel{n_{\alpha 0}} \frac{\cancel{k^2}}{\omega_R^2} \left[1 + 3 \frac{k^2}{\omega_R^2} \frac{k_B T_{\alpha 0}}{m_\alpha} \right] \right\} = \\ &= 1 - \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{\omega_R^2} \cdot \left[1 + 3 \frac{k^2}{\omega_R^2} \frac{k_B T_{\alpha 0}}{m_\alpha} \right] = \\ &= 1 - \sum_{\alpha} \left[\frac{\omega_{p,\alpha}^2}{\omega_R^2} + 3 \frac{\omega_{p,\alpha}^2}{\omega_R^2} \frac{k^2}{\omega_R^2} \frac{k_B T_{\alpha 0}}{m_\alpha} \right] = \\ &= 1 - \sum_{\alpha} \left[\frac{\omega_{p,\alpha}^2}{\omega_R^2} + 3 \frac{\omega_{p,\alpha}^4}{\omega_R^2} \frac{k^2 v_{th,\alpha}}{\omega_R^2 \omega_{p,\alpha}^2} \right] = \\ &= 1 - \sum_{\alpha} \left[\frac{\omega_{p,\alpha}^2}{\omega_R^2} + 3 \frac{\omega_{p,\alpha}^4}{\omega_R^4} k^2 \lambda_{D\alpha}^2 \right] \end{aligned}$$

and finally

$$D_R = 1 - \frac{\omega_{pi}^2}{\omega_R^2} \left(1 + 3 \frac{\omega_{pi}^2}{\omega_R^2} k^2 \lambda_{Di}^2 \right) - \frac{\omega_{pe}^2}{\omega_R^2} \left(1 + 3 \frac{\omega_{pe}^2}{\omega_R^2} k^2 \lambda_{De}^2 \right) \approx$$

$$\approx 1 - \frac{\omega_{pe}^2}{\omega_R^2} - 3 \frac{\omega_{pe}^4}{\omega_R^4} k^2 \lambda_{De}^2 = 0$$

Where we used

$$\frac{\omega_{pi}^2}{\omega_{pe}^2} = \frac{m_e}{m_i} \approx \frac{1}{2000}$$

Now we write it as

$$\omega_R^4 - \omega_{pe}^2 \omega_R^2 - 3 \omega_{pe}^4 k^2 \lambda_{De}^2 = 0$$

and we solve for

$$\omega_R^2 = \frac{1}{2} \left[\omega_{pe}^2 \pm \sqrt{\omega_{pe}^4 + 12 \omega_{pe}^4 k^2 \lambda_{De}^2} \right] =$$

$$= \frac{\omega_{pe}^2}{2} \left[1 \pm \sqrt{1 + 12 k^2 \lambda_{De}^2} \right] \approx$$

$$\approx \frac{\omega_{pe}^2}{2} [1 \pm (1 + 6 k^2 \lambda_{De}^2)]$$

Where in the last passage we used the assumption of

$$k^2 \lambda_{De}^2 \ll 1$$

(we will discuss more in detail about this condition later)

Now, the solution with the minus sign gives an imaginary solution for the real part of ω and is not physical.

The solution with the plus sign gives

$$\omega_R^2 = \frac{\omega_{pe}^2}{2} [2 + 6 k^2 \lambda_{de}^2] =$$

$$= \omega_{pe}^2 [1 + 3 k^2 \lambda_{De}^2]$$

$$\implies \omega_R = \pm \omega_{pe} \sqrt{1 + 3 k^2 \lambda_{De}^2}$$

If we compare it to the isothermal fluid case

$$\omega_R^{(\text{fluid})} = \pm \omega_p \sqrt{1 + k^2 \lambda_{De}^2}$$

or with the adiabatic fluid case

$$\omega_R^{(\text{fluid})} = \pm \omega_p \sqrt{1 + \gamma_E k^2 \lambda_{De}^2}$$

So in kinetic theory we have a $\gamma_E = 3$, this is linked with the degree of freedom.

Having $\gamma_E = 3$ it means that we only have one degree of freedom.

Now, we work with

$$\omega_I = -\frac{D_I(k, \omega_R)}{\partial D_R / \partial \omega_R}$$

with

$$D_R(k, \omega_R) \approx 1 - \frac{\omega_{pe}^2}{\omega_R^2} \left(1 + 3k^2 \lambda_{De}^2 \frac{\omega_{pe}^2}{\omega_R^2} \right)$$

where we assume the second term in the parenthesis as small.

We remember that

$$D_I = -\pi \sum_{\alpha} \frac{\omega_{p,\alpha}^2}{k^2 n_{\alpha 0}} \cdot \frac{\partial F_{\alpha 0}(u)}{\partial u} \bigg|_{\omega_R/k}$$

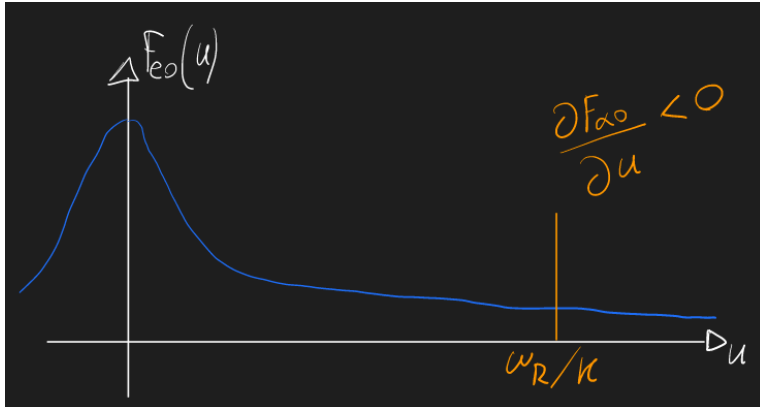
and we evaluate

$$\frac{\partial D_R}{\partial \omega_R} = \frac{\partial}{\partial \omega_R} (-\omega_{pe}^2 \omega_R^{-2}) = 2 \frac{\omega_{pe}^2}{\omega_R^3}$$

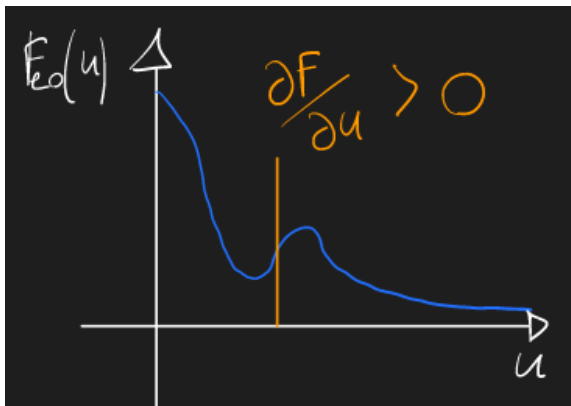
So we can write ω_I as

$$\omega_I = + \frac{\omega_R^3}{2\omega_{pe}^2} \frac{\pi \omega_{pe}^2}{k^2 n_0} \frac{\partial F_{e0}(u)}{\partial u} \bigg|_{\omega_R/k} \propto \frac{\partial F_{e0}(u)}{\partial u} \bigg|_{\omega_R/k}$$

PIC - uno

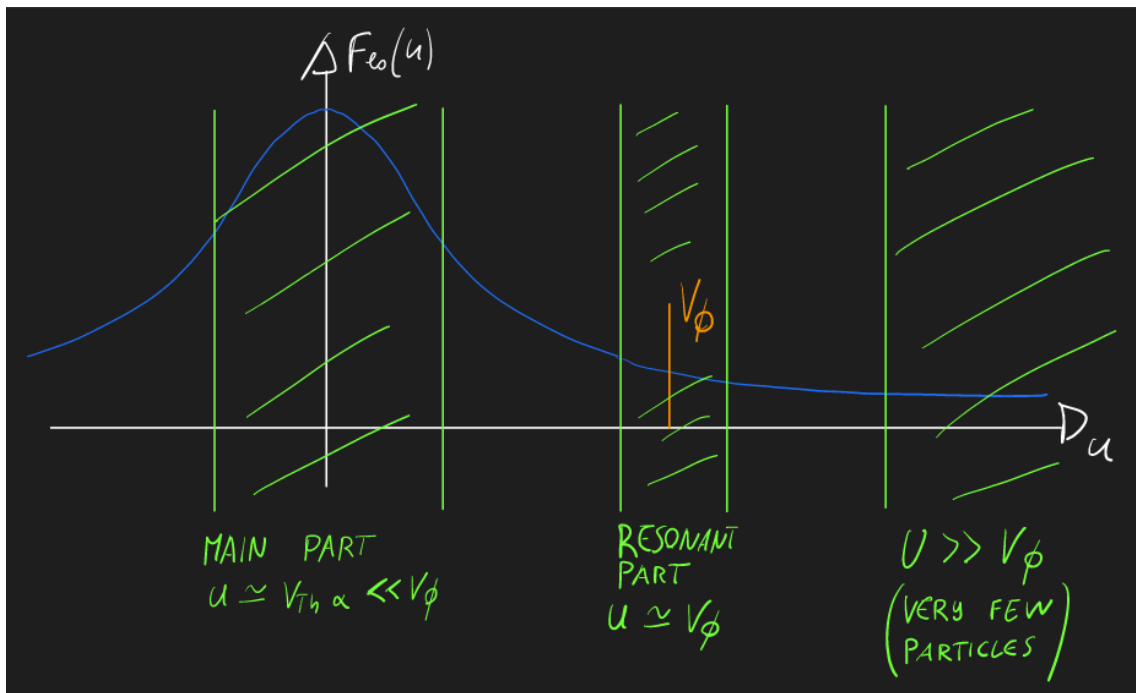


PIC - due

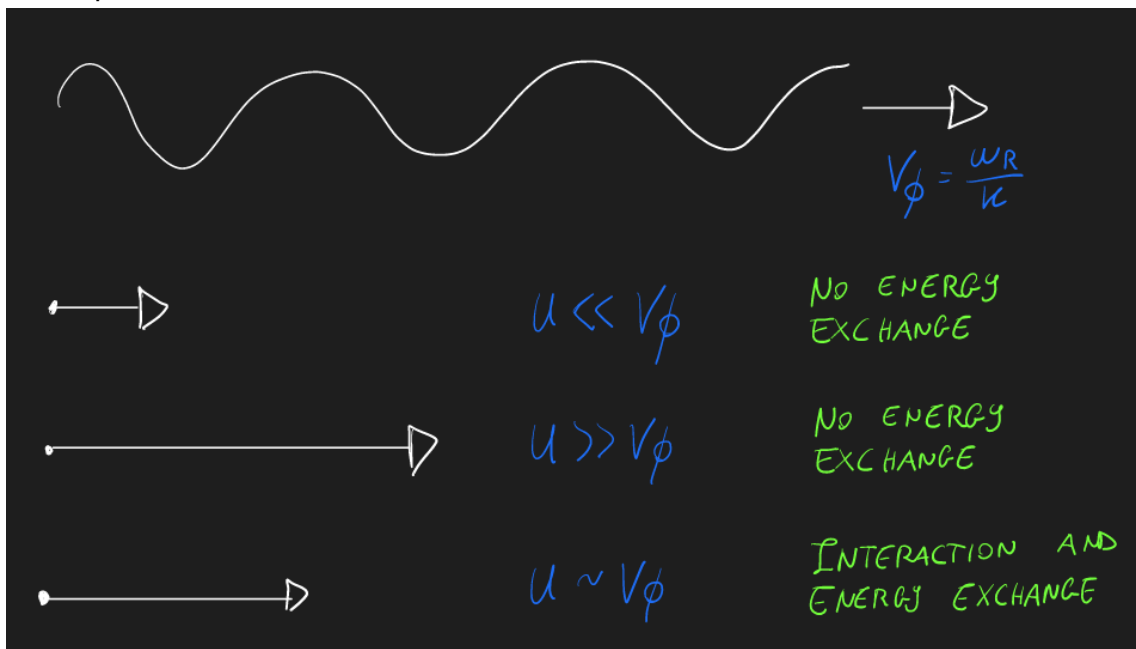


Amplification of waves is common in space where instabilities are born.

PIC - tre



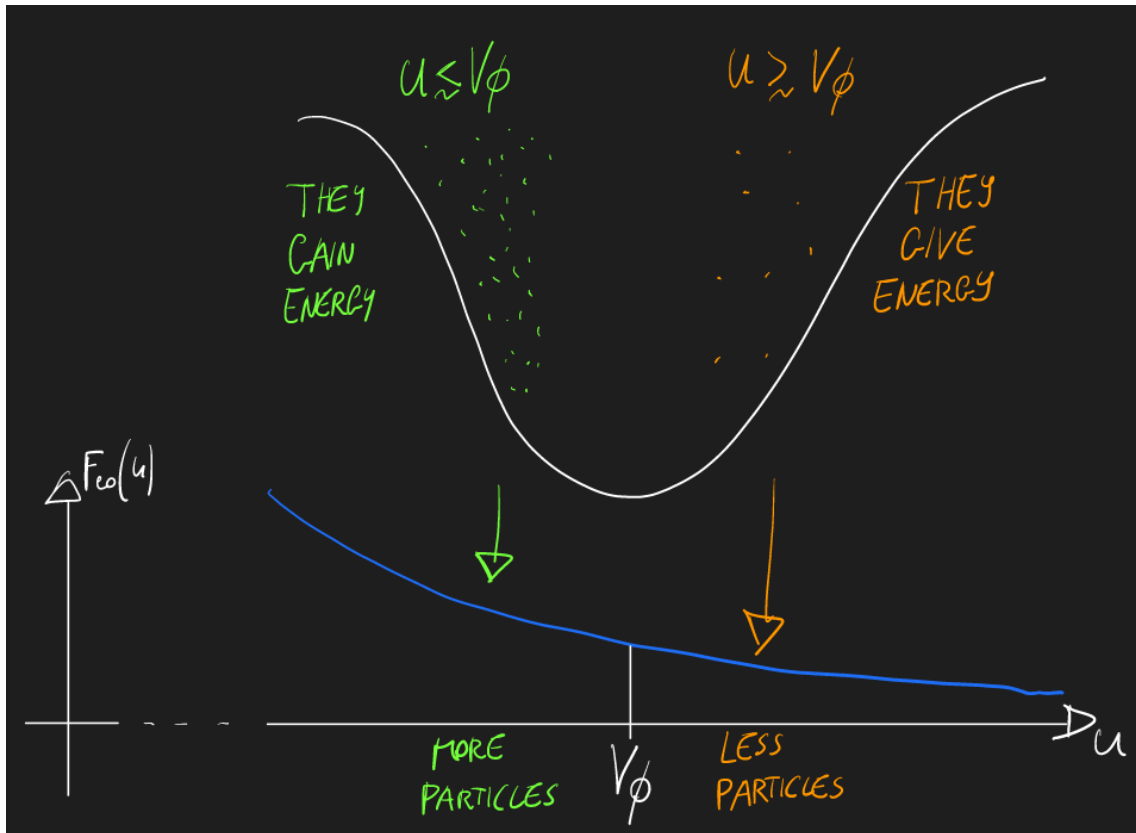
PIC - quattro



u is the velocity of the particle

- $u \ll v_\phi$ or $u \gg v_\phi$, then we don't have energy exchange
- $u \sim v_\phi$, particles feel the electric field generated from the wave for a long time.

PIC cinque e sei (Surf-damping model)



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